

This strategy effectively spends down assets in the same way as the fixed-percentage strategy, and it leads to the largest downside spending reductions, which average from 4 to 6 percent.

Exhibit 6.20

Sustainable Spending Rates from an Investment Portfolio over 30 Years

PAY Rule: Allow for a 10% Chance That Real Wealth Has Fallen Below \$15,000 by Year 30

For a 65-Year-Old Couple, 50/50 Asset Allocation, Rolling 30-Year Retirements

Using SBBI Data, 1926–2016, S&P 500 and Intermediate-Term Government Bonds

Retirement Date Wealth Level = \$100,000

Spending Strategy	Initial Spending Rate	Percentile of Distribution	Real	Real	Real	Real	Change	Year 30	Downside Spending Volatility
			Spending	Spending	Spending	Remaining	in Real	Spending	
			in 10	in 20	in 30	Wealth	Spending	Relative	
			Years	Years	Years	After 30	at Year	to	
						Years	30	Baseline	
Constant Inflation-Adjusted Spending (BASELINE)	4.18%	90th	\$4,180	\$4,180	\$4,180	\$304,500	0%	0%	0%
		50th	\$4,180	\$4,180	\$4,180	\$112,670	0%	0%	0%
		10th	\$4,180	\$4,180	\$4,180	\$14,340	0%	0%	0%
Fixed Percentage Withdrawals	9.49%	90th	\$8,390	\$5,990	\$3,340	\$34,640	-65%	-20%	-3%
		50th	\$5,820	\$3,440	\$2,060	\$20,560	-78%	-51%	-3%
		10th	\$3,590	\$1,880	\$1,400	\$14,950	-85%	-67%	-4%
Endowment Formula—Hybrid	5.45%	90th	\$6,400	\$7,330	\$7,000	\$152,830	28%	67%	-2%
		50th	\$5,170	\$4,620	\$4,230	\$56,710	-22%	1%	-2%
		10th	\$4,120	\$3,580	\$3,190	\$14,510	-41%	-24%	-2%
Endowment Formula—3 Year Moving Average	9.42%	90th	\$8,520	\$5,930	\$3,470	\$34,350	-63%	-17%	-3%
		50th	\$6,370	\$3,610	\$2,100	\$21,130	-78%	-50%	-3%
		10th	\$4,200	\$1,910	\$1,350	\$14,820	-86%	-68%	-3%
Bengen's Dollar Floor-and-Ceiling Withdrawals	4.84%	90th	\$5,810	\$5,810	\$5,810	\$210,440	20%	39%	0%
		50th	\$4,650	\$4,200	\$4,110	\$75,330	-15%	-2%	-1%
		10th	\$4,110	\$4,110	\$4,110	\$12,660	-15%	-2%	-3%